

PEIYU LIU

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SUMMARY

Biostatistics Ph.D. candidate (expected May 2027) specializing in **vaccine and infectious disease trial design** — **cluster-randomized trials**, endpoint selection and validation, and **Bayesian transmission modeling** — with pharmaceutical clinical trial experience at **Bristol Myers Squibb**. Four first-author publications on endpoint efficiency in vector-control trials; author of **vaxineR**, a CRAN package for vaccine-coverage and outbreak-risk analysis. Proven collaborator with **CDC** epidemiologists, clinicians, and virologists. Working knowledge of **CDISC SDTM/ADaM** and TLG workflows.

TECHNICAL SKILLS

Vaccine & ID Trials: Cluster-randomized design, ICC and design effect, endpoint selection and surrogate validation, serological and clinical endpoints, vaccine efficacy and coverage estimation, outbreak-risk and transmission modeling (SIR, household final-size), phylodynamics (BEAST)

Trial Methods: Bayesian hierarchical and adaptive design, MCMC and data augmentation, survival analysis (Cox PH, RMST, Royston-Parmar, AFT), mixed-effects and longitudinal models, missing-data methods (MAR/MNAR sensitivity, multiple imputation), sample size and trial simulation, ICH E9(R1) estimands

Software & Standards: R (Shiny, tidyverse, survival, rstan), Python, SAS, Julia, SQL, Git; CDISC SDTM/ADaM, define.xml, TLG/TFL, protocol and SAP development

PROFESSIONAL EXPERIENCE

University of Florida, Emerging Pathogens Institute — Biostatistics Core
Biostatistician

Gainesville, FL
May 2025 – Present

- Lead biostatistical support across **vaccine studies**, **randomized trials**, and observational **EHR** research, converting investigator questions into **analysis plans**, simulation studies, and reproducible analyses.
- **Authored and released vaxineR on CRAN**, an open-source R package standardizing **vaccination-coverage analysis**, **outbreak-risk estimation**, and intervention simulation, adopted as the analytic basis for a University of Florida faculty white paper.
- **Built posterior-sampling and bootstrap** frameworks for a dengue **cluster-randomized trial**, tracing how **endpoint misclassification** propagates into the estimated intervention effect and establishing the robustness of the primary inference.
- Quantified drivers of pediatric RSV severity in observational **EHR** data using **Cox proportional hazards** and multivariable regression, negotiating clinically interpretable specifications with clinicians and virologists; **peer-reviewed publication**.
- Characterized viral evolutionary dynamics in a US pediatric cohort through **BEAST-based Bayesian phylodynamic** analysis of whole-genome RSV sequences, including prior specification and **MCMC convergence diagnostics**.

Bristol Myers Squibb — Hematology & Cell Therapy Biostatistics
Biostatistics Intern

Remote, U.S.
May 2026 – Aug 2026

- **Designed and deployed an R Shiny clinical utility score** tool collapsing efficacy, safety, and tolerability endpoints into a single **benefit-risk** metric, letting a cross-functional dose team rank candidate doses live in review meetings instead of across successive analysis cycles.
- **Benchmarked FDA Project Optimus**—aligned dose-optimization designs (**PK-BOIN12**, **STEIN12**) across **Phase I/II trial simulations**, reporting **operating characteristics** — correct dose selection rate, overdose exposure, and patients allocation — to drive design selection.
- **Automated raw-to-SDTM/ADaM** variable mapping with **CDISC conformance checks**, cutting manual data preparation and standardizing inputs to endpoint-specific **sample-size calculations**.

University of Florida — Department of Biostatistics
Graduate Research Assistant

Gainesville, FL
Aug 2023 – Present

- **Derived endpoint-specific sample-size formulas** for **cluster-randomized** vector-control trials, linking a **mechanistic exposure model** to **serological**, clinical, and entomological endpoints and proving serological endpoints sharply reduce required trial size in low-incidence settings; **first-author publication** and ASTMH oral presentation.
- **Accelerated Bayesian inference** roughly **10×** by reimplementing a **household final-size transmission model** in **Julia**, after extending it to incomplete households via **data augmentation** and partnering with **CDC** epidemiologists to separate household from community transmission.

- **Formulated a Bayesian latent-process** model reconstructing seasonal exposure from **longitudinal serology**, jointly estimating latent exposure histories, **antibody kinetics**, and between-subject heterogeneity.
- Established which parameters stay **estimable** when subject identity is masked across repeat sampling — a constraint shared by field and **decentralized studies** — and built **robust estimation** procedures validated on three field cohorts (Benin, Thailand).
- **Validated phylogeny-informed transmission estimators** against epidemiologic-only estimators across **1,000 stochastic SIR simulations**, integrating survival and **transmission-tree likelihoods**.

Hunan Institute of Science and Technology — Department of Mathematics

Changsha, China

Research Fellow and Lecturer

Jun 2021 – Aug 2023

- Directed a **\$7,500 funded research project** applying **statistical learning** methods (**EM, Lasso, GAMs**) and spatial analysis to multinational COVID-19 data across 10 countries.
- Lectured on undergraduate probability and statistics and mentored modeling competition teams to a **CSIAM Meritorious Mentor Award** (2022).

EDUCATION

Ph.D., Biostatistics — University of Florida, College of Public Health and Health Professions

Expected May 2027

GPA 3.9/4.0. Coursework: Analytical Methods for Infectious Diseases, Survival Analysis, Causal Inference, Longitudinal Analysis, Statistical Machine Learning, Biostatistical Methods and Computing.

M.S., Computational Mathematics — Shaanxi Normal University, Xi'an, China (top 5%)

May 2021

B.S., Statistics — Shaanxi Normal University, Xi'an, China (top 5%)

May 2017

SOFTWARE & DEPLOYED APPLICATIONS

vaxineR (CRAN) — R package for **vaccine-coverage analysis** and **outbreak-risk modeling** — cran.r-project.org/package=vaxineR

Dengue Trial Endpoint Comparison — **Serological** versus **clinical endpoint** efficiency for **cluster-randomized trials** —

peiyuliu.shinyapps.io/anpp-trial-comparison

SIR Model Estimation — Least squares, bootstrap, and **MCMC** estimation of transmission parameters — peiyul.shinyapps.io/sir-estimation-comparison

Sample Size Suite — CDISC-aligned dataset checks and sample size calculation across endpoint types —

peiyuliu.shinyapps.io/clinical-data-ssize-suite

SELECTED PUBLICATIONS

Full list: scholar.google.com/citations?user=O6KFoOUAAAAJ

1. **Liu, P.**, Rong, L., Cummings, D., & Hitchings, M. (2026). Mosquito salivary protein antibodies as efficient endpoints for vector control trials: a mechanistic comparison with clinical and entomological outcomes. *Infectious Disease Modelling*, 11(4), 1368–1386.
2. **Liu, P.**, et al. (2026). Practical identifiability and Bayesian inference of seasonal exposure from serology data. Under review.
3. Santos, L. A., Castillo, A., Gustani-Buss, E., Hitchings, M. D. T., **Liu, P.**, et al. (2026). Genomic diversity of human respiratory syncytial virus type A in a US pediatric cohort after the onset of the COVID-19 pandemic. *Microorganisms*, 14(8), 1656.
4. **Liu, P.**, He, S., Rong, L., & Tang, S. (2020). The effect of control measures on COVID-19 transmission in Italy: comparison with Guangdong province, China. *Infectious Diseases of Poverty*, 9(130), 1–13.
5. **Liu, P.**, et al. Bayesian household transmission modeling with incomplete data. In preparation.

SELECTED PRESENTATIONS & AWARDS

Oral presentation, **ASTMH Annual Meeting**, Washington, DC (Nov 2026); oral presentation, UF Public Health and Health Professions Research Days (Apr 2026).

PhD Travel Award, ASTMH (2026) · PhD Student Travel Award, SISIMID (2025) · Meritorious Mentor Award, CSIAM (2022) · National Scholarship, Ministry of Education, China (2020).